FLORIN CUCONASU

Rome, Italy
Personal Website | florin.cuconasu10@gmail.com | LinkedIn | GitHub | Scholar

EDUCATION

NATIONAL PHD IN ARTIFICIAL INTELLIGENCE — *Nov 2023 – (Nov 2026)*
1st year PhD Student at Sapienza University of Rome — *Rome, Italy*
- Best Poster Award at the M2L Summer School

MS IN ENGINEERING IN COMPUTER SCIENCE — *Sep 2021 – Oct 2023*
Sapienza University of Rome — *Rome, Italy*
- 110/110 cum laude & Honors Program

BS IN INFORMATION ENGINEERING — *Sep 2018 – Oct 2021*
Sapienza University of Rome — *Rome, Italy*
- 110/110 cum laude & Honor Mention

PUBLICATIONS

A Tale of Trust and Accuracy: Base vs. Instruct LLMs in RAG Systems [paper] — *ArXiv, 2024*
The Power of Noise: Redefining Retrieval for RAG Systems [paper] [interview] — *SIGIR, 2024*
RRAML: Reinforced Retrieval Augmented Machine Learning [paper] — *AIxIA Conference, 2023*

EXPERIENCE

SAPIENZA UNIVERSITY OF ROME — *Feb 2024 – May 2024*
Teacher Assistant — *Rome, Italy*
- Taught 12 hours of lectures to 100+ students in a Programming Techniques course.
- Collaborated with the lead professor to design and develop programming exercises and exam materials.

META — *Dec 2023 – Dec 2023*
Workshop – Personal Aria and Digital Data Timeline — *Menlo Park, CA*
- Tested Aria glasses at Meta HQ, exploring integration with user data through hands-on sessions.
- Collaborated in a 5-person hackathon team to develop a ChatGPT-based chatbot.
- Designed a RAG system analyzing Aria-recorded menus and user data for personalized food suggestions.

PEER-TO-PEER EDUCATION — *Sep 2016 – Aug 2018*
- Tutored 5 high school students in Math and Physics, resulting in a 25% increase in their mean test scores.

PROJECTS

E-commerce – Microservices Architecture | *React, HTML5, CSS3, Django, PostgreSQL, Docker, Scrum, Jira*
- Led a team of 5 to build a microservices-based e-commerce application.
- Designed REST APIs in Django for efficient communication between microservices.
- Conducted Scrum meetings and managed user stories and project goals on Jira.

TV Movie Analysis – Interactive Visualization Tool | *D3.js, Pandas, SciPy, Sklearn*
- Led a team of 3 to design and develop an interactive visualization tool for analyzing TV movie scheduling.
- Integrated data from different sources, including web scraping, resulting in a dataset of 100k tuples.
- Designed customized distance functions for Multidimensional Scaling (MDS) to plot data points in 2D.

Coreference Resolution | *PyTorch, Pandas*
- Developed a BERT-based model for Pronoun Coreference Resolution.
- Achieved top class accuracy (88.2%) on Google's GAP dataset, while reducing training time by 3x.

ABOUT ME

Programming Python • SQL • PyTorch • Weights & Biases • Pandas • Sklearn • Git
Languages English (*Proficient*) • Italian (*Native*) • Romanian (*Intermediate*)
Interests NLP • Generative AI • Information Retrieval • Retrieval Augmented Generation • RL